

Teacher Guide for Kindergarten

Standard

K.LS3.1 Make observations to describe that young plants and animals resemble their parents.

Tennessee Academic Standards for Science: Page 19

Three-dimensional Learning Performance for Lesson

Students will ask descriptive questions after observing photographs of young organisms and their parents* in order to show that young plants and animals resemble their parents** highlighting the patterns of features used to determine if creatures are related based on similarities and differences.***

Science and Engineering Practice for Lesson

*Asking Questions**

Students will be shown images of young animals and plants and then images of parent animals and plants, with the purpose of asking and refining scientific questions about the images. Initially, students will most likely ask broad questions, such as “Why do some animals look different as babies?” The teacher will then guide students to make more careful observations and notice particular features that could be evidence of a relationship between a young animal and their parent. The goal is to have students generate a more scientific line of questioning. For example, one student may be guided to ask, “Why does a baby chick have yellow feathers, but a grown-up chicken has brown/black/white feathers?” While another student may ask, “Why are their feathers different colors but their mouths/beaks look similar?” By refining their questions to focus more on similar and different features, students make more refined observations about which features provide evidence that young plants and animals resemble their parents.

Disciplinary Core Idea for Lesson

*Life Science 3: Heredity***

“Organisms have characteristics that can be similar or different. Young animals are very much, but not exactly, like their parents and also resemble other animals of the same kind. Plants also are very much, but not exactly, like their parents and resemble other plants of the same kind.”

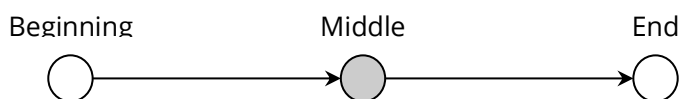
Crosscutting Concept for Lesson

*Patterns****

Students observe that plants and animals of a certain species display a pattern of similar features or characteristics. These patterns are typically even more similar when observing organisms that are part of the same family.

Prior Knowledge

Location Within Instructional Unit



- Concepts taught prior to this lesson:
 - We use our senses to notice things, such as features on a plant or animal
 - Young animals often resemble their parents
 - We determine this by looking for how similar their features are to each other

Materials

- [Tree Detective Sheet](#)
- [Animal Matching Cards](#)
- [“Animal Parents and Kids” YouTube slideshow](#)
- [Image of Mature Pine Tree](#)
- [Image of Pine Tree Sapling](#)
- [Image of Detective \(Cartoon\)](#)
- [Image of Kid Detective](#)
- Optional “Detective” Lens (hand lens) for students

Lesson Sequence Outline

1. Review that young animals often resemble their parents
2. Complete matching animal activity
3. Guide students to ask a similar question about plants (e.g., “Do young plants resemble their plant parents?” or “Do all living things resemble their parents?”)
4. Discuss how a detective is a special kind of policeman that investigates things to get all the possible clues, which they then use to solve a case or answer a question

5. Set up students for activity by having them get their 'detective lens' out so they can think like a detective
6. Have students engage with the *Tree Detective Sheet* and collect for formative assessment

Instructional Notes

The teacher begins the lesson by reviewing from a previous lesson with students that young animals often resemble their parents. To do this, the teacher (re)plays the YouTube slideshow "[Animal Parents and Kids](#)," and pauses the video at each slide so that they can call on students to identify similar features that provide evidence that these animals are related. For example, students may say "The parent porcupine and baby porcupine both have spiky backs!" or "The puppy has the same color brown fur as its parent." A focus for this part of the lesson is to have students generate evidence that describes how the animals are similar, rather than just declaring "They look the same!"

Once the slideshow is over, the teacher will have students demonstrate their understanding of animals and their parents by having them match the parent animal with its kid using the *Animal Matching Cards*. The cards should be copied, cut out ahead of time, and laminated if possible. Students can work in pairs to match their cards. Whenever a student matches animals together, they should describe evidence for why they think those animals are related. For example, the student may say, "I think the cat and the kitten are related because they have the same grey colored fur." Again, the focus for this activity is to have students work on identifying specific features that are similar between animals.

After students have completed the *Animal Matching Cards* activity, the teacher gathers students back together and tells them a story. In the story, the teacher was out walking in nature when they noticed a big pine tree. The teacher displays the [Image of Mature Pine Tree](#) to students so that they can understand what the tree looks like. Then, the teacher continues saying that during that same walk, they noticed another tree that looks similar in some ways, but different in other ways. The teacher displays the [Image of Pine Tree Sapling](#) to students and asks students if they notice what is similar about these two pictures. As students offer suggestions, such as, "They both have pine needles!" the teacher pretends to do some deep thinking before saying to students, "You guys are noticing some really interesting similarities between these two trees. Hearing all your thoughts is causing me to have some questions about these trees now. Does anyone else have a question or is maybe wondering something about these two trees?" From here, the teacher guides

students to ask questions about the trees with the goal of having students wonder if the trees could possibly be related.*

**Note: The goal of having students ask questions about the trees is to hopefully have a student ask if the two trees are possibly related, like the animals in the matching game and on the slideshow. Students may require some scaffolding to arrive at this question, so the teacher can guide the class to ask these sorts of questions by suggesting things like "I notice several of you are asking questions about how one tree is little and one tree is really big. When we looked at animals and their parents on the slideshow earlier, was one of them always big and one of them always little? (Yes!) What could this mean about the big tree and the little tree then? Are you saying that they could be related?"*

Once students are primed to examine if the trees are possibly related to each other, the teacher tells students, "Trees can be harder to figure out than animals. This is because trees in some ways are very different than animals, but if we go searching for clues very carefully, we might find some evidence that helps us find out the real answer." The teacher then displays the [Image of Detective \(Cartoon\)](#) and asks students if they know what a detective does. If they do not know, the teacher explains that a detective is someone who looks very carefully for clues to help them solve something. The teacher explains that a detective is usually shown with a hand-lens because they use tools to help them look for clues. The teacher emphasizes that most importantly, though, a detective uses those clues to ask "Detective Questions." These are questions related to the mystery that the detective is trying to solve. The teacher can give an example of this to students, such as "For example, when I was walking on the nature trail the other day and saw those two trees I thought to myself 'I noticed those two trees look like a parent tree and a baby tree because they have the same type of needle-leaves, but one was really big and the other one was really small. I wonder if these trees are related like the animals we saw earlier?'"

The teacher can model this further by working with the class to point out any other similarities that they notice that might be evidence that two trees are related. Some similarities include: The leaves appear the same, they both have leaves growing from a branch, they both are growing up out of the ground, they both have green leaves and a brown trunk, etc. This may need to go on for a few minutes to give students a chance to notice and describe similarities between the two trees.

When students are ready to move on, the teacher informs them that for this next activity, the students will need to become detectives to solve the mystery of whether the following trees could be related. In order to do this, they must look for evidence (similar features) in the pictures on the activity sheet they will receive. This means they will need to use their

“Detective Hand-lens” to make careful observations of some images of trees and determine how the trees are similar. The teacher can display the [Image of Kid Detective](#) to help solidify this point. In order to work as “tree detectives,” the students will each receive a copy of the [Tree Detective Sheet](#) as well as an optional hand-lens. Students will need to use the sheet to do the following:

1. Page 1
 - A. Look closely at two images of trees to notice how they are similar and how they are different
 - i. Students can use their ‘detective hand-lens’ to help them focus on the images and hone in on specific features of the trees
 - B. Describe one or two ways that the trees are similar
 - C. Describe one or two ways that the trees are different
2. Page 2
 - A. Use the images of two “mystery” plants to notice how they are similar and how they are different
 - B. Formulate their own “detective question” about the two plants
 - i. Examples:
 1. Are the plants related?
 2. Will the smaller/“baby” plant grow up to be like the bigger/“parent” plant?
 3. Will the “arms”/branches on the smaller plant grow?
 4. Do the plants have the same parts?

As students are working, the teacher is monitoring students and assisting as needed. Once finished, the teacher can collect the sheets to look over student responses as a formative assessment of student understanding.

Student Responses and Formative Assessment

Young Birch/Mature Birch observations:

- Similarities include: trunks are both white-ish, both types of trees grow out of the ground, both trees have leaves, etc.
- Differences include: Young birch trunks are more skinny than mature birch trunks, mature birch has more leaves than young birch, mature birch trunks form a ‘Y’ shape and young birch trunks go up in a straight line, etc.

Mystery Desert Plants

The images are of a young Saguaro cactus and a mature Saguaro cactus.

Students should notice how similar the two plants appear in the pictures. The branches on the young Saguaro will develop over time to look more like the mature Saguaro.

As students are deciding on a 'detective' question related to these plants, they should choose the question that asks if the plants come from the same family based on their similarities.

Citations and Resources

Image of Pine Tree:

<https://www.flickr.com/photos/forestfortrees/4359236961>

Image of Pine Tree Sapling:

<https://www.flickr.com/photos/96713863@N00/1290126678/>

Image of Detective (Cartoon):

<https://pixabay.com/en/inspector-man-detective-male-160143/>

Image of Kid Detective:

<https://pixabay.com/en/detective-magnifying-glass-viewing-788592/>

Animal Matching Cards

Image of Goat:

<https://pixabay.com/en/goat-billy-goat-cute-goat-2644998/>

Image of Kid:

https://commons.wikimedia.org/wiki/File:Hausziege_04.jpg

Image of Deer:

<https://pixnio.com/fauna-animals/deers/deer-female-in-wild>

Image of Fawn:

<https://www.flickr.com/photos/slopjop/828388505>

Image of Kangaroo:

https://en.wikipedia.org/wiki/File:Kangaroo-in-flight_cropped.jpg

Image of Joey:

https://commons.wikimedia.org/wiki/File:Macropus_rufogriseus_rufogriseus_Joey_1.jpg

Image of Cat:

<https://pxhere.com/en/photo/986828>

Image of Kitten:

<https://pixabay.com/en/cat-tomcat-kitty-kitten-2866167/>

Image of Duck:

https://commons.wikimedia.org/wiki/File:White_domesticated_duck_stretching.jpg

Image of Duckling:

<https://pxhere.com/en/photo/994760>

Image of Dog:

<https://pixabay.com/en/dog-labrador-light-brown-pet-1210559/>

Image of Puppy:

<https://commons.wikimedia.org/wiki/File:Too-cute-doggone-it-video-playlist.jpg>

Tree Detective Sheet

Image of Mature Birch Tree retrieved from:

<https://pixabay.com/en/nature-outdoors-woods-birch-tree-3070475/>

Image of Young Birch Tree retrieved from:

https://commons.wikimedia.org/wiki/File: Birch_forest_with_birds_nest_Gullmarsskogen.jpg

Image of Walnut Tree retrieved from:

<https://www.flickr.com/photos/brighton/4621671958>

Image of Young Saguaro Cactus retrieved from:

<https://www.nps.gov/tont/learn/nature/saguaro.htm>

Image of Mature Saguaro Cactus retrieved from:

<https://www.nps.gov/orpi/learn/nature/saguaro-cactus.htm>